

# VIDAA: Delivering a multilingual smart TV voice assistant with Dialogflow

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To offer its customers a smart TV control system designed for smart TVs around the world, VIDAA leverages Dialogflow to quickly develop a smart TV voice control capability supporting multiple languages with 90% detection accuracy.



## About VIDAA

Founded in 2019 and headquartered in Atlanta, USA, VIDAA provides a smart TV operating system and content platform to many leading TV brands around the world. Its self-developed TV OS, which was released in early 2020, boasts wide content selection and a truly lean-back user experience. In mid-2021, VIDAA added a self-developed smart voice control system to its smart TV OS.

Industries: Technology

## Google Cloud results

- Helps develop a smart TV voice control system supporting 13 languages in just six months
- Simplifies machine learning model training process to realize advanced voice control features with Dialogflow
- Expands support of new languages quickly with Multilingual

Location: US, China

## More than 90% accuracy in voice intent

Google Cloud (<https://cloud.google.com/>)

Dialogflow Enterprise Edition  
(<https://cloud.google.com/dialogflow-enterprise/>)

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## Agent on Dialogflow

As smart TVs become increasingly popular around the world, more and more people rely on voice assistants to control their TVs. Besides basic controls like volume adjustment and channel switching, consumers also expect voice assistants on smart TVs to have more advanced capabilities exclusively designed for TVs, such as brightness adjustment and content search.

VIDAA (<https://www.vidaa.com/>) has been dedicated to meeting this demand. Founded in 2019, VIDAA aims to build a market-leading smart TV platform that delivers the best possible user experience. In early 2020, it released VIDAA Smart TV OS, its self-developed smart TV operating system (OS), which is now used in more than 100 countries.

“The users of  
our smart TV OS  
live across  
different  
continents, so  
we need a voice  
assistant  
development

tool that  
supports all the  
most commonly  
used languages,  
especially in  
Southeast Asia,  
and Dialogflow  
is the only one  
that meets our  
needs.”

—Rajin Persuad, VP  
Product, VIDAA

At first, VIDAA integrated its smart TV OS with various third-party voice assistants. But since these voice assistants were designed for general use, VIDAA wanted to expand the capabilities and supported languages of its smart TV OS . To offer its international customers a voice control system designed exclusively for smart TVs, VIDAA decided to build a TV voice assistant on its own. In early 2021, the company adopted Dialogflow (/dialogflow) to develop its voice control system because of the variety of languages that the tool supports, and the ever-advancing voice recognition technology of Google Cloud (<https://cloud.google.com/>).

“The users of our smart TV OS live across different continents, so we need a voice assistant development tool that supports all the most

commonly used languages, especially in Southeast Asia, and Dialogflow is the only one that meets our needs,” recalls Rajin Persuad, VP product at VIDAA. “Moreover, the proven commitment of Google Cloud to keep advancing its voice recognition technology makes us convinced that we can continue improving our voice control system with the help of Google Cloud.”

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Product, VIDAA

## **Developing a TV voice assistant with 13 languages in six months**

VIDAA's development team leverages the machine learning (ML) models on Dialogflow to develop its smart voice control system. To create a voice control feature, the team only needs to collect and input a dozen of conversation texts in different languages, without worrying about ML model management. Persuad says that compared with building ML models from scratch, using Dialogflow for voice assistant development has significantly reduced the time and effort required.

The clear documentation and intuitive interface design of Dialogflow have also helped enhance VIDAA's development efficiency. The company's engineers can always find detailed explanations in documentation easily, which prevents them from making unnecessary mistakes. It only took them a short time to get the hang of the ML model training system on Dialogflow. As a result, VIDAA successfully developed a TV voice assistant supporting 13 languages in only six months with a small development team.

"The easy-to-use nature of Dialogflow and the simplified ML model training process that it provides are truly beneficial for us, because we wanted to launch our TV voice assistant as soon as possible," Persuad notes. "With the 13 languages that our

voice control system initially supported, we were already able to offer a TV voice assistant to users in many countries where there was no similar product available.”

## **A voice control system with advanced features and 93% intent detection accuracy**

Since the launch of its TV voice assistant, VIDAA has expanded its voice control capabilities from basic TV commands to advanced features like content search by names of actors or film release years. When its development team was building more complex voice control features, it needed to train the ML models with more text elements to ensure that its voice assistant understands users’ intents correctly. At one point, the number of text elements the VIDAA team wanted to input reached the limit of the Dialogflow system. With technical advice given by the Google Cloud team, VIDAA’s engineers were able to bypass the limit and complete the development of more advanced features with a high level of intent detection accuracy.

The fact that Dialogflow doesn’t need phrases to be an exact match to understand users’ intents also helps increase the intent detection capability of VIDAA’s TV voice assistant. Overall, the company’s voice control system can understand a user’s intent with an accuracy rate more than 93%.

“Ensuring that our TV voice assistant supports a high intent-detection accuracy rate is essential for

us to provide excellent user experience. With Dialogflow, we not only enjoy its great intent-detection capability, but also continue improving the accuracy rate ourselves when building more advanced features,” says Persuad.

“As the voice recognition technology of Google Cloud keeps advancing, and the number of languages supported by Dialogflow continues to grow, we believe that we have a good foundation to enlarge our user base and offer even better voice control services in the future.”

*—Rajin Persuad, VP  
Product, VIDAA*

## **Continuous expansion of supported languages and voice control features**

VIDAA continues to extend the supported languages and features of its TV voice assistant on Dialogflow. With the Multilingual Agent feature of Dialogflow, the VIDAA team is able to add additional languages to the already trained ML models, instead of training ML models all over again in new languages. This has helped the team quickly increase the number of the supported languages of its voice control system from 13 to more than 20. Moving forward, the team also plans to broaden the search capability of its TV voice assistant by allowing users to look for content by more types of keywords.

Persuad says, “Thanks to Dialogflow, we were able to build a mature voice control system in a short period of time. As the voice recognition technology of Google Cloud keeps advancing, and the number of languages supported by Dialogflow continues to grow, we believe that we have a good foundation to enlarge our user base and offer even better voice control services in the future.”